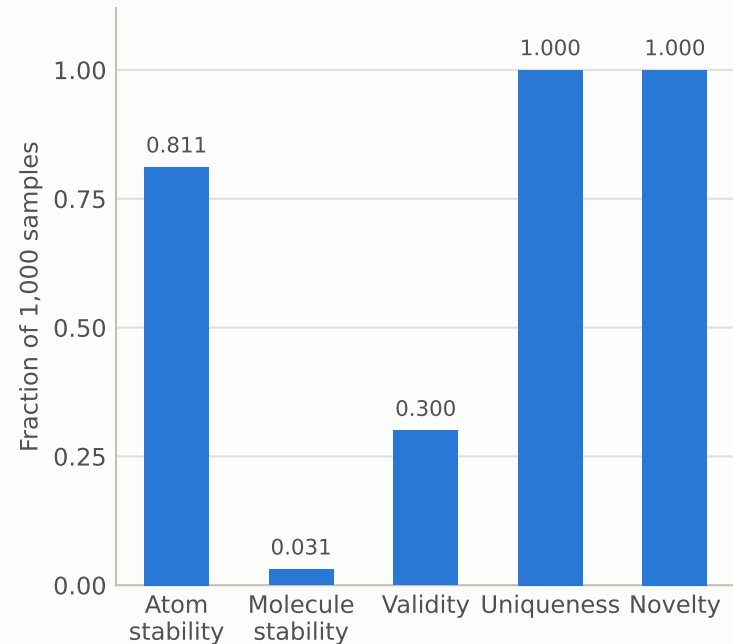
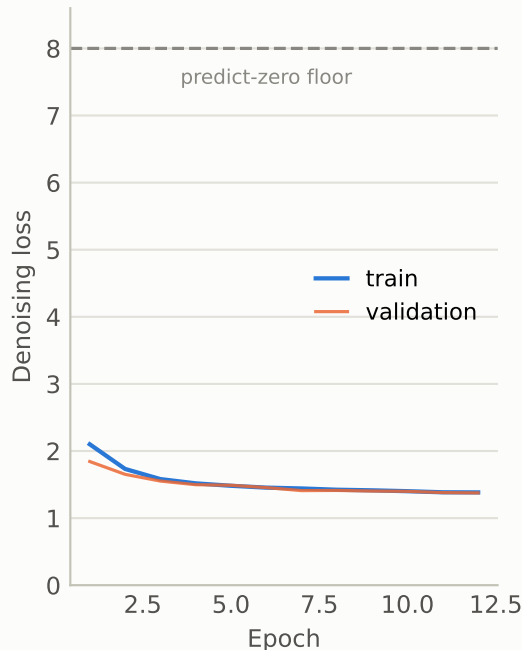


# QM9 unconditional generation: EGNN denoiser

## Sample quality, EGNN denoiser on QM9



## Training converged



Model setup - EGNN

```
denoiser      egnn, 464,393 params
layers x width 4 x 128
data          QM9, 130,831 molecules
split         80 / 10 / 10, seed 0
diffusion     DDPM, T=500, polynomial_2
parameterised epsilon-prediction, L2
training      12 epochs, batch 64
optimiser     AdamW, lr 0.0002, EMA 0.999
gauge         none (equivariant denoiser)
bonded fraction 1.000 (QM9 data 1.000)
```

Atoms are bonded at data-like distances, so validity here is a genuine measurement, not the single-atom artefact seen with the MLP arm.